

VecScaleDB ANN Benchmark Report

Distributed Vector Database Evaluation

This report compares VecScaleDB against the standard ANN benchmark datasets used in the Milvus evaluation setup: SIFT10K, SIFT1M, and Deep1M. The focus is the Figure 10b-style QPS-vs-nodes experiment, with secondary metrics for latency, errors, data coverage, and benchmark readiness.

Generated from local workspace metrics

Timestamp: 2026-05-04T22:57:49.697918+00:00

System	VecScaleDB
Repository	https://github.com/sanchit-22/Distributed-Vector-DB
Team	Deadlock and Chill
Member	Sanchit Kumar - 2024201042
Member	Priyanshu Jha - 2024201062
Member	Rohini Mandlekar - 2025901004
Index	Faiss IVF_FLAT
Distance metric	L2
Primary scaling artifact	QPS vs reader nodes
Reader modes	Replicated round-robin and scatter-gather
Durability	WAL + MemTable + immutable segments

Report contents

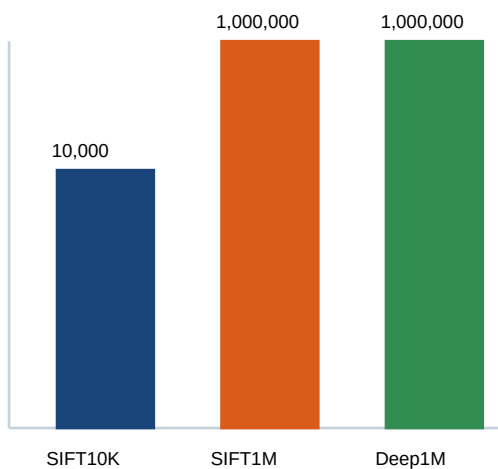
- Dataset specification and local availability
- Project architecture overview
- SIFT1M QPS and latency charts from local scaling_results.json
- Cross-dataset comparison framework for SIFT10K, SIFT1M, and Deep1M
- Methodology and commands to regenerate missing measurements

Dataset Comparison

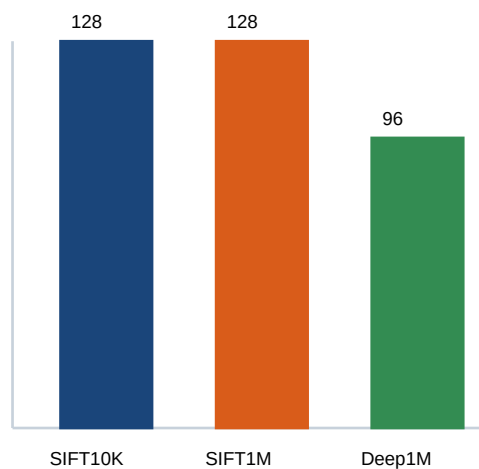
All three datasets are public ANN benchmarks. SIFT10K is used for fast correctness and recall checks, SIFT1M is the primary local performance benchmark, and Deep1M is included as the paper-replication target when the 96-dimensional dataset is supplied locally.

Dataset	Vectors	Dims	Metric	Available	Role
SIFT10K	10,000	128	L2	yes	Unit tests and correctness checks
SIFT1M	1,000,000	128	L2	yes	Primary performance benchmark
Deep1M	1,000,000	96	L2	yes	Paper replication dataset

Dataset scale, log10(vectors)



Vector dimensions

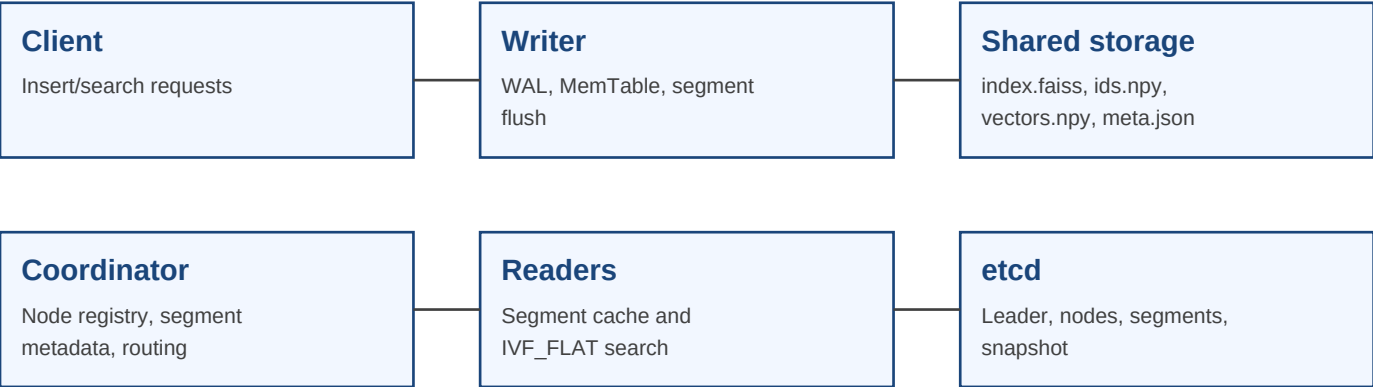


Local dataset inspection

- SIFT10K: available; path=data/sift1m/sift1m.hdf5; size=500.8 MB
 - distances: shape=[10000, 100], dtype=float32
 - neighbors: shape=[10000, 100], dtype=int32
 - test: shape=[10000, 128], dtype=float32
 - train: shape=[1000000, 128], dtype=float32
- SIFT1M: available; path=data/sift1m/sift1m.hdf5; size=500.8 MB
 - distances: shape=[10000, 100], dtype=float32
 - neighbors: shape=[10000, 100], dtype=int32
 - test: shape=[10000, 128], dtype=float32
 - train: shape=[1000000, 128], dtype=float32
- Deep1M: available; path=data/deep1m/deep1m.hdf5; size=3669.75 MB
 - distances: shape=[10000, 100], dtype=float32
 - neighbors: shape=[10000, 100], dtype=int32
 - test: shape=[10000, 96], dtype=float32
 - train: shape=[9990000, 96], dtype=float32

Project Overview

VecScaleDB uses a writer/readers/coordinator split. Inserts go to the writer, which appends to the WAL, stores recent writes in a MemTable, and flushes immutable Faiss-backed segments to shared storage. Readers load those segments and answer search requests. Coordinators store metadata in etcd and route search requests.



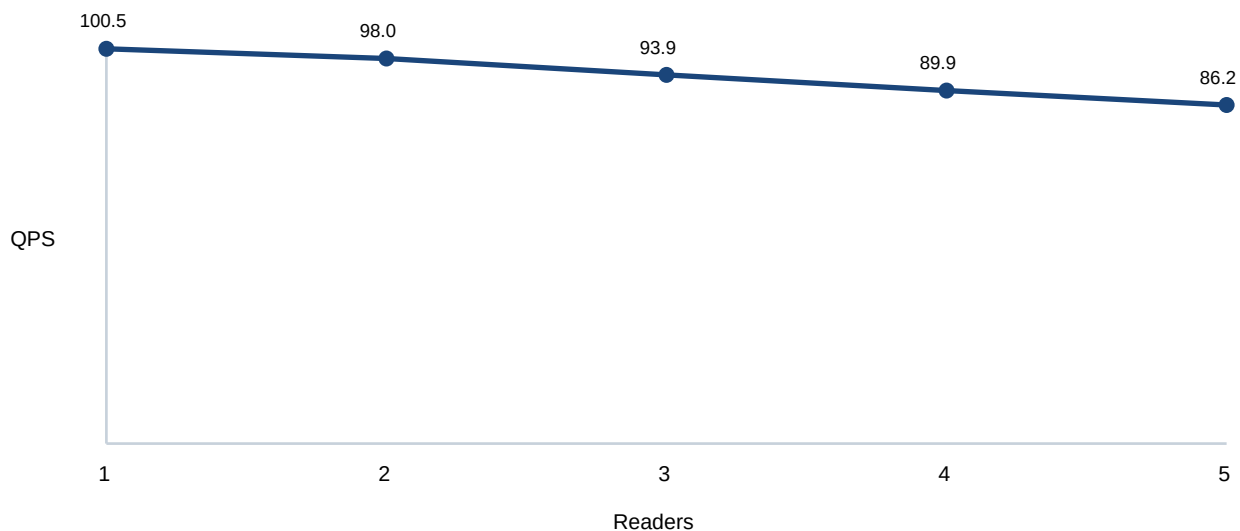
Feature coverage

Feature	Implementation
Distributed architecture	3 coordinators, 1 writer, 5 readers, shared storage, etcd
Load balancing	replicated readers receive coordinator round-robin search traffic
Vector search	Faiss IVF_FLAT with configurable nprobe
Dynamic data	insert/delete through WAL-backed writer pipeline
Snapshot isolation	search uses snapshot_id derived from WAL/segment LSNs
Fault tolerance	readers reload segments; writer replays WAL; coordinator leader failover

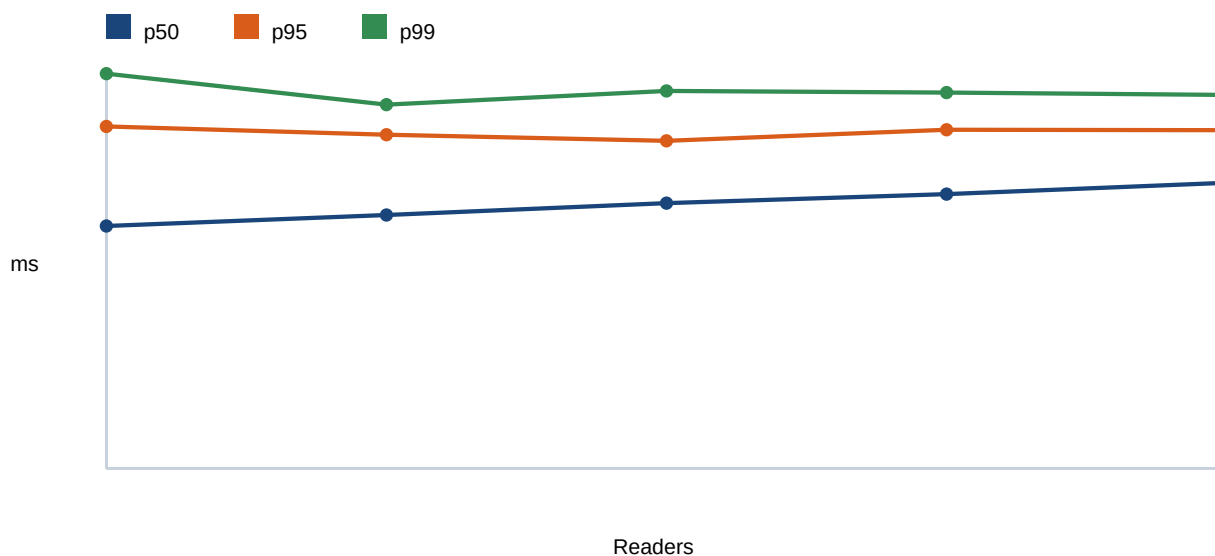
SIFT1M QPS Scaling

The current local scaling input comes from scaling_results.json. It measures coordinator search throughput while varying active reader count from 1 to 5. The expected Figure 10b-style artifact is a QPS-vs-reader-count curve.

QPS vs reader nodes



Latency percentiles vs reader nodes



Readers	QPS	p50	p95	p99	Errors	Efficiency
1	100.50	315.61	445.19	513.91	0	100.0%
2	98.03	329.92	434.42	473.58	0	48.8%
3	93.86	345.37	426.48	491.42	0	31.1%
4	89.87	357.09	440.86	489.28	0	22.4%
5	86.20	371.87	440.33	486.23	0	17.2%

Cross-Dataset Evaluation Matrix

The report separates dataset readiness from benchmark readiness. A dataset can be available locally even if a full scaling or recall run has not been executed yet. This keeps the comparison honest and reproducible.

Dataset	Data	Best QPS	p95	Recall	Load	Status
SIFT10K	yes	109.77	357.88 ms	R@10=0.104, R@100=0.025	9331/s	qps measured
SIFT1M	yes	100.50 @ 1 readers	445.19 ms	R@10=0.998, R@100=0.996	9703/s	measured
Deep1M	yes	104.18 @ 1 readers	422.43 ms	skipped	7713/s	measured

Dataset and benchmark coverage

Data	yes	yes	yes
Scaling	no	yes	yes
	SIFT10K	SIFT1M	Deep1M

Interpretation

- SIFT10K validates correctness quickly and should report Recall@10/Recall@100.
- SIFT1M is the primary performance dataset and currently has local QPS scaling data.
- Deep1M is listed for paper replication but requires a local 96D HDF5 file and loader validation.
- The current SIFT1M run is not near-linear; QPS decreases as readers increase, which suggests local resource contention or benchmark setup effects that should be investigated before claiming positive scaling.

Methodology And Reproduction

The report is generated from local files, not hard-coded benchmark claims. Run the commands below to refresh data, then rerun the report scripts.

Collect metrics

```
python report/collect_metrics.py
```

Generate PDF

```
python report/generate_report.py
```

SIFT1M scaling

```
python -m vecscaledb.bench.scaling_eval --dataset data/sift1m/sift1m.hdf5
--coordinator-url http://127.0.0.1:8000 --reader-counts 1 2 3 4 5 --duration 60
--concurrency 32 --query-limit 1000 --output
report/results/sift1m_scaling_results.json --auto-scale
```

SIFT10K recall

```
python -m vecscaledb.bench.recall_eval --host 127.0.0.1 --port 8000 --dataset
data/sift1m/sift1m.hdf5
```

Metrics used

- QPS: completed search requests divided by elapsed seconds.
- Latency p50/p95/p99: percentile search latency in milliseconds.
- Errors: failed or incomplete search requests in the benchmark window.
- Scaling efficiency: observed speedup divided by reader-count increase.
- Recall@k: overlap between returned top-k IDs and ground-truth nearest neighbors.

Caveats

- Milvus published values are not embedded numerically in this repo; the comparison reproduces the metric shape and dataset setup.
- Deep1M is marked pending until the local dataset is supplied.
- A single laptop Docker run can show resource contention, so final claims should be made from controlled repeated runs.